

Lower Urinary Tract Cancer



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KEYWORDS

• Transitional cell carcinoma • Prostate tumor • Chemotherapy • Radiation therapy

KEY POINTS

- Transitional cell carcinoma (TCC) is the most common urinary tract tumor in both cats and dogs.
- The most common location in the bladder for canine TCC is the trigone, making local control challenging. Local control is also challenging for canine prostate tumors.
- Systemic therapy is often the treatment of choice for urinary tract tumors and may be used in combination with other local therapies.
- Median survival time (MST) for dogs with TCC treated with nonsteroidal anti-inflammatory drugs (NSAIDs) alone is 4 to 6 months and with NSAIDs plus chemotherapy is 9 to 11 months.
- MST for cats with TCC treated with surgery, chemotherapy, NSAIDs, or a combination of these modalities is approximately 8.5 to 12 months.

CANINE URINARY BLADDER TUMORS

Urinary bladder tumors account for only about 2% of all canine tumors. Of these, transitional cell carcinoma (TCC) is the most common, affecting approximately 20,000 dogs each year.^{1–4} Canine TCC develops most often in the trigone region of the bladder and commonly also involves the urethra or prostate. Interestingly, canine TCC closely resembles high-grade, muscle-invasive TCC in humans, making it a relevant animal model for research and translational study.^{1,3}

Patient Evaluation Overview

A number of risk factors associated with an increased risk for development canine urinary bladder TCC have been noted (**Table 1**). In addition to the inherent risk factors outlined in **Table 1**, obesity and exposure to certain chemicals (older flea control

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Table 1 Inherent risk factors for canine urinary bladder TCC	
Factor	Odds Ratio
Sex (female:male)	1.71–1.95:1
Breed	
Scottish Terrier	18.09
Shetland Sheepdog	4.46
Beagle	4.15
Wire-Haired Fox Terrier	3.20
West Highland White Terrier	3.03

Data from Refs.^{2–9}

products, herbicides, and pesticides) are also considered to increase the risk of TCC; newer topical flea/tick control products, specifically those containing fipronil, are not of concern.^{2–9} Therefore, dogs at highest risk are obese female dogs exposed to insecticides (odds ratio, 28). To date, no studies evaluating prevention in high-risk breeds have been performed, but given the data available, keeping pets at a lean body weight, limiting exposure to herbicides and pesticides, and feeding vegetables at least three times per week seem to be appropriate as preventive measures.^{2–9}

Clinical Signs

Clinical signs (eg, hematuria, stranguria, pollakiuria, and dysuria) mimic other lower urinary tract diseases, and many dogs have concurrent urinary tract infections (UTIs), which may initially delay diagnosis. The secondary UTI is often resolved with appropriate antibiotic therapy and may result in temporary alleviation of the clinical signs, but signs recur. Thus, the presentation of a middle-aged dog with its first UTI should prompt imaging of the bladder to rule out bladder stones and/or a bladder mass in addition to ruling out other medical conditions that may predispose to a UTI (such as diabetes or hyperadrenocorticism). Physical examination may reveal a thickened urethra or trigonal region or an irregular prostate, which may be painful on digital rectal examination. Some cases may reveal no abnormalities on physical examination.⁵ Only about 1 in 6 cases of urinary bladder TCC will have metastasis documented at the time of diagnosis, usually to the lymph nodes and/or lungs.^{2,3,5,10,11} Mortality associated with TCC most commonly occurs owing to urinary tract obstruction or clinical signs associated with the primary tumor, rather than metastasis, owing to the challenges with local control.

Diagnostic Tests

Imaging

A urethral or bladder mass on ultrasound is suggestive for TCC or other neoplasia, but polyps and polypoid or other chronic cystitis can mimic neoplastic conditions, so ultimately cytology or histopathology is required. Thoracic and abdominal imaging with both radiography and ultrasonography in patients with known or suspected TCC is recommended to screen for metastatic disease, with common locations including lymph nodes, lungs, and liver. Less frequently, TCC may metastasize to other sites, including bone. Recently, CT has been shown to be the optimal way to serially image and monitor bladder tumors.¹² Dogs with higher stage tumors (eg, those with larger tumors or with metastasis) have shorter expected survival times.⁷

Antigen tests

Urine bladder tumor antigen tests have been reported to be 88% sensitive for detection of TCC, but a high number of false-positive results limit their usefulness.^{13,14} These tests are useful only in a clinically normal dog with no known urinary tract abnormalities (such as screening for at-risk breeds); if the test is negative, there is a very low probability of TCC.

Definitive diagnosis

Options for obtaining a sample of a bladder mass include biopsy via cystoscopy, cystotomy, traumatic catheterization, or fine needle aspiration (FNA). Evaluation of sediment cytology samples is often misleading because urinary bladder epithelial cells may mimic a neoplastic population when cystitis is present. Bladder mass FNA is controversial because seeding of these tumors along the needle tract has been reported.^{15,16} However, FNA may be considered in cases in which the tumor is large and nonresectable, after counseling of the owners on the risks and careful consideration of all of the options available for diagnosis. A complete blood count and full serum chemistry profile with electrolytes to evaluate overall health and biochemical function is recommended along with urinalysis and urine culture. Azotemia may be present in patients with any degree of urinary tract obstruction.

Treatment Options

Surgery

Complete excision of TCC is often not feasible in dogs owing to urethral, prostatic or ureteral involvement, as well as the morbidity associated with complete cystectomy. In dogs where excision is feasible (eg, apically located tumors), surgery may prolong survival.^{17,18} Robat and colleagues¹⁸ reported a significantly increased overall median survival time (MST) in dogs who underwent chemotherapy plus cytoreductive surgery compared with dogs who received chemotherapy alone (217 vs 133 days, respectively; $P = .02$). Molnar and colleagues¹⁷ found a significant survival benefit in a small number of dogs who underwent both surgery and chemotherapy compared with dogs who received chemotherapy alone (475 vs 31 days, respectively; $P < .001$). A “field effect” has been proposed for TCC because the entire bladder epithelium is exposed to carcinogens in the urine, and thus multifocal malignant changes are likely present. Therefore, recurrence either at the site of original surgery owing to inability to take wide margins, or at distant sites in the bladder owing to the field effect, is common. Nonetheless, given the available data, consideration should be given to surgery followed by chemotherapy when excision is possible.

Stenting

Urethral or ureteral stents can be used in dogs with evidence of obstruction and are often placed by fluoroscopic or ultrasonographic guidance.^{19,20} Female dogs are commonly incontinent after urethral stent placement, which raises concerns for potential owner exposure in dogs that go on to receive chemotherapy. Approximately 39% of dogs (male and female) are incontinent after stent placement, and other complications include reobstruction (22%), stent migration, and bladder atony.¹⁹ Other options for relief of ureteral or urethral obstruction include urinary diversion via subcutaneous ureteral bypass devices. It is unknown at the time of this writing how these new procedures may affect survival times, but because obstruction is often the factor leading to euthanasia in patients with TCC, any method that can manage obstruction would likely prolong survival. Laser ablation has also been described recently and may provide relief of clinical signs but has not yet been thoroughly studied or shown to alter overall survival greatly.^{21,22}

Chemotherapy

Systemic chemotherapy is often the standard-of-care treatment for TCC owing to the lack of good options for local control. Typical approaches include treatment with nonsteroidal antiinflammatory drugs (NSAIDs) alone or in combination with systemic chemotherapy agents. Piroxicam is commonly favored as the first-line NSAID, although treatment with other NSAIDs such as deracoxib and firocoxib has also been reported.^{23–25} Complete responses have been reported in a small number of cases treated with piroxicam alone, with an MST of 5.9 months.²⁴ The MST in a single report of 26 deracoxib-treated dogs was longer; however, those dogs received other chemotherapy agents after deracoxib.²⁵ Piroxicam has been noted to cause both nephrotoxicosis as well as gastrointestinal upset; thus, routine monitoring of the dog's renal function and gastrointestinal signs is indicated when administering piroxicam. Misoprostol can be used to prevent NSAID-induced ulceration; however, in the authors' experience, many dogs experience gastrointestinal cramping and upset from this drug. Misoprostol is also an abortifacient and, thus, women of child-bearing age should not be exposed to this drug. Drugs such as omeprazole or famotidine could be considered to reduce the risk of gastrointestinal side effects from NSAIDs; although there is no published evidence in companion animals, there is evidence in humans that this approach can reduce the risk of NSAID-induced gastric or duodenal ulceration.^{26–28} Routine biochemical monitoring for renal and liver function and evidence of gastrointestinal ulceration is recommended for all dogs on NSAIDs, especially those considered at risk of urinary tract obstruction.

Cytotoxic agents with reported activity include mitoxantrone, carboplatin, cisplatin, doxorubicin, vinblastine, gemcitabine, vinorelbine, metronomic chlorambucil, and intravesical mitomycin C.^{18,29–39} Mitoxantrone, carboplatin, and vinblastine in combination with NSAIDs are common first-line agents with reported response rates ranging from 35% to 38%.^{29,31,35} Although favorable tumor responses to cisplatin and piroxicam are also noted, this combination is not recommended because of high nephrotoxicity.^{32,33} A summary of study results for treatments for canine TCC is shown in **Table 2**. Overall, it seems there are similar outcomes for many drugs, and thus the advantages and disadvantages of each—cost, time, toxicity, and systemic health of the pet, including renal and hepatic function—should be taken into consideration when choosing a treatment regimen.

Radiation therapy

Early reports of radiation therapy for TCC in dogs were discouraging owing to side effects; however, more recently published protocols have been well-tolerated. Invasive TCC in humans is often treated with radiation therapy, either alone or in combination with surgery and/or chemotherapy, and radiation has been shown to be the most important component of therapy for this disease in humans.^{40–44} To date, radiation therapy has not been extensively studied in dogs. High dose per fraction (21.88–27.00 Gy) intraoperative radiotherapy had a high risk of complications in one study, with 46% of dogs experiencing increased frequency of urination, 46% urinary incontinence, 38% cystitis, and 15% stranguria.⁴⁵ Despite this, MST was 16.8 months, implying that radiation therapy may have a role in treatment of canine TCC.

In 2004, Poirier and colleagues⁴⁶ investigated the use of piroxicam, mitoxantrone, and a coarse-fractionated protocol of radiation therapy (6 weekly fractions of 5.75 Gy) in 10 dogs with TCC, but did not find a superior response rate when compared with another study of mitoxantrone and piroxicam alone (response rate of 22% vs 34.5%; respectively).²⁹ Despite the lower objective response rate, 90% of patients had amelioration of clinical signs, and late side effects occurred in 3

Table 2
Summary of study results for treatments for canine urinary bladder TCC

Modality	Availability	Side Effects	Outcome	Recommended?
NSAIDs ^{23–25}	GP	Nephrotoxicity, hepatotoxicity	MST, ~4–6 mo	Yes, unless other contraindications (eg, renal failure)
Cytotoxic chemotherapy ^{23–25,29–39}	Specialist recommended/ some GPs may be trained in chemotherapy	Different drugs carry different risks and side effects: bone marrow suppression, GI upset, other; in general, ~80%–85% of dogs tolerate chemotherapy with a good quality of life	MST, ~8–11 mo	Yes, unless other contraindications (eg, biochemical abnormalities, MDR1 mutations)
Surgery ^{17,19}	Specialist, some GPs	General risks of surgery and anesthesia; TCC is highly prone to transplantation so care should be taken to reduce that risk	In combination with chemotherapy: MST, 7.1–15.5 m; more study needed to define outcome	Careful case selection recommended; apically located tumors are likely best candidates for surgical resection
Radiation therapy–palliative ^{29,46}	Specialist	Pollakuria, urinary incontinence, cystitis, and stranguria	Improved clinical signs in 90% of cases in 1 report	Consider in advanced cases
Radiation therapy–IMRT ⁴⁷	Specialist	Acute: hematuria and stranguria; late: strictures (rectal, ureteral, and urethral)	Limited data (single report) in combination with chemotherapy MST, 21.4 mo	Consider if no distant metastatic disease and committed owners; considerable time and financial commitment
Stenting ^{19,20}	Specialist	Incontinence (39%), reobstruction (22%), stent migration, and bladder atony; if incontinent, raises concerns for home and owner exposure to chemotherapy if chemotherapy is pursued after stent placement	Resolves obstruction; survival times not well-described, but expected to increase outcomes; about 20% reobstruct	Consider if obstructed; careful owner counseling recommended before pursuing stenting in case of incontinence after the procedure
Laser ablation ^{21,22}	Specialist	Stranguria, hematuria, stenosis, seeding of TCC, urethral and bladder perforation, bacterial cystitis	More study needed but reported MST 9.8–13.4 mo in combination with other therapies	Consider if obstructed; cases should be carefully selected

Abbreviations: GI, gastrointestinal; GP, general practitioner; IMRT, intensity-modulated radiation therapy; MST, median survival time; NSAID, nonsteroidal anti-inflammatory drug; TCC, transitional cell carcinoma.

dogs. More recently, image-guided, intensity-modulated radiation therapy (IMRT) has been studied in dogs with genitourinary tract tumors (bladder, urethra, and prostate).⁴⁷ In 21 dogs treated with IMRT (54–58 Gy delivered in 20 fractions on a Monday to Friday basis) and chemotherapy, MST was 654 days.⁴⁷ Two patients developed acute urinary complications, including hematuria and stranguria (1 each). Four patients developed urethral, ureteral, or rectal strictures as late complications; all were successfully palliated with either stenting or surgery.⁴⁷

Evaluation of Outcome and Long-Term Recommendations

Overall, canine TCC carries a poor prognosis, and the majority of dogs succumb to this disease. However, the quality of life for most dogs receiving chemotherapy is excellent, and new therapies are being evaluated constantly. Dogs with larger tumors invading locoregionally or with evidence of metastasis (ie, more advanced stage) have a less favorable prognosis with shorter survival times or shorter time to obstruction.^{2,7,47} Dogs at an increased risk for metastasis include younger dogs (increased risk of lymph node metastasis), dogs with prostate involvement (increased risk of distant metastasis), and dogs with larger tumors (increased risk for both nodal and distant metastasis).^{2,47}

Many publications describe heavily pretreated dogs or dogs that go on to receive other treatments, which makes survival data for dogs with this tumor challenging to summarize. However, the following deductions may be made from the published data (**Table 2**):

- With NSAID therapy alone, expected MST is about 4 to 6 months.
- Addition of chemotherapy to NSAID therapy is expected to increase MST to approximately 9 to 11 months.
- Surgery, stenting, and radiation therapy all warrant consideration in appropriate cases and may improve survival.

CANINE PROSTATE TUMORS

Canine prostate tumors are uncommon, occurring in less than 1% of dogs.⁴⁸ The majority are carcinomas and may arise from the glandular epithelium, prostatic ducts, or prostatic urethra. Other reported prostate tumors in dogs include lymphoma,^{49,50} hemangiosarcoma,^{51,52} and leiomyosarcoma.⁵³

Patient Evaluation Overview

Risk factors

Older, neutered male dogs are predisposed (**Box 1**). Many predisposed breeds are also predisposed to urinary bladder TCC, which may reflect the fact that many prostate tumors are TCCs arising from the prostatic urethra.

Clinical signs are indistinguishable from other causes of prostatic disease and most commonly include tenesmus, constipation, dyschezia, hematuria, stranguria, and dysuria, as well as systemic signs such as lethargy, weight loss, and anorexia. Patients may also present with lameness or skeletal pain owing to bony reaction or metastatic disease, and signs of skeletal metastasis may be the only presenting sign in some cases.^{54–58}

Physical Examination

Prostatomegaly (typically asymmetric) is the most common abnormality, although it may not be present in all dogs.⁵⁶ No other consistent physical examination abnormalities have been reported, but may include abdominal pain, a palpable abdominal

Box 1**Potential risk factors for canine prostate tumors**

Age^{54,55,99,100}:

- Older dogs (median, 9–10 years)

Neuter status^{48,99–101}:

- Neutered dogs at increased risk (OR 2–8 times compared with intact dogs, depending on type of tumor; most effect in TCC and least in adenocarcinoma)

Breed^{48,102}:

- Mixed-breed
- Shetland Sheepdog
- Scottish Terrier
- Beagle
- German Shorthaired Pointer
- Airedale
- Norwegian Elkhounds
- Bouvier des Flandres

Abbreviations: OR, odds ratio; TCC, transitional cell carcinoma.

mass, pyrexia, depression, and musculoskeletal pain; special attention should be paid to signs of pain or lameness given the high propensity for bony metastasis. Prostatomegaly and associated clinical signs in an older neutered male dog should raise a strong suspicion for prostate cancer.

Diagnostic Tests

Radiography

Prostatomegaly and mineralization are associated strongly with neoplasia in neutered dogs. More caution is needed in interpretation in intact dogs (also seen with benign prostatic hyperplasia, prostatitis, and prostatic cysts).^{59,60} Other abnormalities noted with prostatic neoplasia include bony reaction, especially of the pelvis or lumbar vertebrae, and lymphadenopathy.^{59,60} Thoracic radiographs should be performed in every case of prostatic neoplasia because the lungs are common sites of metastasis.^{55,56,58} Skeletal metastases are common (especially affecting the axial skeleton) and most, but not all, dogs with skeletal metastases also had visceral metastases in 1 study.⁵⁵ Skeletal survey radiographs could be considered if aggressive therapy is an option (eg, definitive radiation therapy), but at the very least, radiographs of affected areas for dogs displaying signs of musculoskeletal pain or myelopathy should be performed.

Abdominal ultrasound

Disruption of the prostatic capsule, focal to multifocal areas of increased echogenicity/mineralization in the prostate, and enlarged lymph nodes are associated with neoplasia.^{59,61} A complete abdominal ultrasound should be performed to evaluate for metastasis—local lymph nodes are the most common sites, but others include the liver, kidneys, distant lymph nodes, large intestines, heart, adrenal glands, brain, and spleen. Diffuse peritoneal metastasis/carcinomatosis is also reported.^{55,56,62}

Definitive diagnosis

Cytology samples may be obtained via prostatic wash or FNA. Occasionally, neoplastic cells may be detected in the urine.⁵⁶ Biopsies for histopathology may be obtained via ultrasound-guided needle core biopsy or laparotomy. Given that there are currently no proven prognostic or therapeutic differences between different types of carcinoma (eg, ductal vs glandular origin vs TCC of prostatic urethra), the authors do not recommend additional procedures for histopathology in cases where cytology confirms carcinoma. In cases of suspected prostate cancer, the risk of seeding cancer cells along a needle tract should be considered before aspiration or biopsy,^{15,63} and this is of particular concern when owners may consider aggressive therapy (eg, IMRT); noninvasive options such as prostatic wash are preferable in these situations. Urine culture should be performed in all patients because secondary infections are common.⁵⁶

Treatment Options

Non-steroidal anti-inflammatory drugs

Canine prostate tumors frequently express cyclooxygenase (COX)-1 and COX-2,^{64–66} and the use of NSAIDs (piroxicam or carprofen) has shown a significant improvement in survival in dogs with prostate tumors (most of which were of urothelial origin; [Table 3](#)).⁶⁶ Cytotoxic chemotherapy has not been evaluated specifically in canine prostate tumors, but its utility in canine bladder TCC suggests that it could be considered, and anecdotally it is recommended commonly. There is some in vitro evidence for the effectiveness of paclitaxel,⁶⁷ and this may bear further clinical investigation given the recent approval of Paccal Vet-CA1. However, the current conditional approval means that this drug cannot be used routinely for canine prostate cancer (not a labeled indication).

Radiation Therapy

The previously mentioned study of IMRT showed promising outcomes for dogs with prostatic neoplasia. However, this modality is not suitable for dogs with distant metastatic disease, and owners should be counseled about risks of side effects, as discussed.⁴⁷ Palliative radiation therapy may be used to relieve severe clinical signs, urethral obstruction, or pain from bone metastases. The risk of late side effects to the gastrointestinal and urinary tract is theoretically considerable with palliative protocols, but clinically not of great concern because of the palliative intent—most patients do not live long enough to experience late side effects.

Other

Urethral stenting can palliate dogs with obstruction, and although incontinence is common, most dogs were judged to have good outcomes in 1 study.²⁰ Based on the available literature, NSAIDs should be recommended in all patients, and IMRT should be offered to suitable candidates; cytotoxic chemotherapy should be considered, but evidence as to its true efficacy is lacking. Surgery is not recommended owing to a high incidence of complications and lack of improvement in outcomes over NSAIDs alone, and other investigational treatments such as photodynamic therapy have likewise shown no additional benefit.

Evaluation of Outcome and Long-term Recommendations

Canine prostate cancer has a poor to grave prognosis given the difficulty in controlling the local disease and the high risk of metastasis. With treatment, reported MST range from approximately 3 to 4 months to up to 1.8 years for dogs with disease

Table 3
Treatment options in canine prostate cancer

Modality	Availability	Side Effects	Outcomes	Recommended?
NSAIDs ⁶⁶	GP	Renal, GI	MST, 6.9 vs 0.7 mo for untreated	Yes, unless other contraindications
Cytotoxic chemotherapy	Specialist/some GPs	Bone marrow suppression, GI upset, other	Unknown in primary prostate tumors	Consider owing to high metastatic potential, although unknown benefit; indicated in lymphoma
Radiation therapy, palliative	Specialist	Acute: colitis most common; late: usually not clinically relevant owing to short survival	Anecdotal palliation of clinical signs, no large studies	Consider if obstruction, clinical signs not responding to other therapy, bone metastasis
Radiation therapy, definitive IMRT ($\pm$ chemotherapy) ⁴⁷	Specialist	Colitis, skin effects most common—generally mild, severe late side effects possible but rare	Potential clinical benefit in 90%; median event-free survival (local recurrence or metastasis), 10 mo	Consider if no distant metastatic disease and committed owners, considerable time and financial commitment
Urethral stent ²⁰	Specialist	Incontinence, stranguria	5/7 dogs judged to have good to excellent outcome	Consider if obstructed, palliative only
Photodynamic therapy ($\pm$ surgery, NSAIDs) ^{94,95}	Specialist/investigational	Not identified	No apparent benefit overall, one report of potential long-term response	Not recommended
Total prostatectomy ⁹⁶	Specialist	Incontinence, hematuria, detrusor atony, others	Complications life-limiting in many dogs	Not recommended
Subcapsular prostatectomy (laser or surgical) $\pm$ NSAIDs, adjuvant therapy ^{97,98}	Specialist	Complications requiring euthanasia possible ($\leq 26\%$), otherwise mild, eg, incontinence	Improvement in clinical signs in most, but MST ~ 3 –6 mo	Not recommended

Abbreviations: GI, gastrointestinal; GP, general practitioner; IMRT, image-guided, intensity-modulated radiation therapy; MST, median survival time; NSAID, nonsteroidal anti-inflammatory drug; TCC, transitional cell carcinoma.

confined to the prostate treated with IMRT.⁴⁷ Therapy is generally aimed at relieving clinical signs and improving comfort with long-term tumor control not expected in most cases.

FELINE LOWER URINARY TRACT TUMORS

Lower urinary tract tumors in cats are very uncommon and mostly affect the urinary bladder, with primary urethral and prostatic tumors being exceedingly rare.^{68–74} Like dogs, the most common lower urinary tract tumor is TCC.^{75–80} Others include various sarcomas,^{75,81,82} leiomyomas,⁸³ lipomas,⁸⁴ and lymphoma, either as part of multicentric disease or localized to the urinary bladder.^{75,77,78,85,86}

Patient Evaluation Overview

Signalment

Older cats (median age, 10–15 years) are predisposed to urinary bladder tumors in general, but lymphoma may be seen in cats as young as 1 year of age.^{75,76,78–80} A male predisposition has been seen in some studies but not others,^{76,78,80} and there is no known breed predilection. Interestingly, a genetic predisposition is possible with 2 case series describing urinary bladder TCC in related captive fishing cats, though there are no similar reports in related domestic cats.^{87,88}

Clinical signs

Clinical signs are consistent with lower urinary tract disease (Box 2), and systemic signs are also common.^{69,79,83–86} Rarely, TCC is diagnosed in cats presenting for non-urinary-tract-related signs.^{76,78,80} Because the median age of cats with urinary bladder neoplasia is significantly older than cats with idiopathic lower urinary tract disease,⁸⁹ lower urinary tract signs in any older cat should raise the suspicion for neoplasia; cats with severe clinical signs (eg, marked hematuria) or cats with chronic intermittent lower urinary tract signs also bear further investigation for an underlying cause. Some cats with urinary bladder neoplasia have a history of suspected or confirmed idiopathic cystitis for months to years,^{78,80,83} and chronic inflammation may have promoted tumor development in these cases (although an association has not been demonstrated).

Physical Examination

There are no findings that are specific for urinary bladder neoplasia, although a distended bladder or abdominal mass effect may be palpable. An enlarged, irregular

Box 2

Clinical signs of lower urinary tract neoplasia in cats

- Hematuria (may be profound)
- Stranguria/dysuria/pollakiuria
- Incontinence
- Inappropriate elimination
- Urethral obstruction
- Systemic signs (eg, anorexia, weight loss, lethargy)

Modified from Refs. ^{76,78–80,86}

prostate on rectal examination is very suspicious for prostatic neoplasia; other prostatic disease is even less common than prostatic neoplasia in cats.

Diagnostic Tests

Diagnostic tests are aimed at diagnosing the primary tumor and screening for metastasis. Although metastasis is not common (0%–27%) at diagnosis in TCC,^{76,80} widespread disease can occur (lymph nodes and lungs are common sites), and complete abdominal ultrasound and thoracic radiographs should be performed.^{76–78,86,90}

Imaging

Reported abnormalities on abdominal radiographs (plain or contrast studies) include bladder distension, thickened or calcified bladder wall, and presence of a mass.⁷⁸ Ultrasound has mostly superseded radiography for detection of lower urinary tract neoplasia. A mass is identifiable in most cases, but the bladder wall may seem to be thickened diffusely.⁸⁰ In contrast with dogs, TCC occurs in cats most commonly in nontrigonal regions of the bladder wall.^{76,78–80} Cats with tumors causing urethral or ureteral obstruction may have ultrasonographic evidence of hydronephrosis.

Definitive diagnosis has been achieved via urine cytology,^{76,80,86} cytology of catheter suction biopsies/traumatic catheterization,^{76,80} FNA,^{76,91} and histopathology, usually via laparotomy and cystotomy for biopsy or partial cystectomy.^{78–80} Although not reported in cats, there may be a risk of tumor seeding to the abdominal wall along needle tracts, and less invasive methods (eg, traumatic catheterization) should be considered first, especially for surgical candidates.

Common abnormalities on routine laboratory tests (complete blood count, chemistry panel, and urinalysis) include anemia and azotemia, either owing to preexisting chronic kidney disease or obstruction caused by the tumor.^{75,76,78} In cats with urinary bladder lymphoma, feline leukemia virus (with or without feline immunodeficiency virus) testing should be performed.⁸⁵ There is 1 report of suspected paraneoplastic hypereosinophilia in a cat with urinary bladder TCC and widespread metastasis.⁹⁰ Hematuria and pyuria are common, and neoplastic cells in the urine have been identified in rare cases,^{76,80} but atypical transitional cells in urine, as discussed in canine TCC, should always be interpreted with caution. Urine culture is recommended in all cases of suspected urinary tract neoplasia because secondary infections are common.^{76,78}

Treatment Options

Urinary bladder TCCs in cats frequently express COX-1 and COX-2,^{80,88} so NSAIDs are a rational therapeutic choice. Surgical excision is commonly performed for TCC and may also be effective for other urinary bladder tumors and some prostate tumors.^{72,73} Urethral stents or cystostomy tubes may be used to palliate patients with urethral obstruction.^{80,92,93} **Table 4** for details about these options.

Evaluation of Outcome and Long-term Recommendations

Many cats with urinary bladder TCC are euthanized at diagnosis owing to debilitation, urethral obstruction, or evidence of metastatic disease. Ultimately, most cats with urinary bladder TCC will die owing to local or metastatic disease. Local recurrence is common after surgical excision, but because prolonged tumor control may be seen in some cases and surgery may improve survival, surgery is recommended for accessible (nontrigonal), nonmetastatic TCC.^{76,80} Median survival time for cats treated with surgery, chemotherapy, NSAIDs, or a combination of these modalities is approximately 8.5 to 12 months.^{76,80} Given the paucity of the literature, it is impossible to

Table 4 Treatment options for cats with urinary bladder neoplasia			
Modality	Availability	Side Effects	Recommended?
Surgery (partial cystectomy) ^{72,73,76,80}	GP	Decreased bladder capacity	Yes, for tumors other than lymphoma in accessible (nontrigonal) locations, without metastatic disease
NSAIDs: Meloxicam (0.1 mg/kg PO SID initially, then 0.05 mg/kg SID) or piroxicam (0.3 mg/kg PO SID 2–3 times per week). Although not evaluated, robenacoxib should be considered owing to its favorable safety profile. All off-label. ^{76,80}	GP	Renal failure, GI (diarrhea, melena)	Yes, in TCC—can palliate clinical signs (potentially for months) used alone, consider adjunctively after surgery. Preexisting chronic kidney disease does not necessarily preclude NSAID use, but renal function must be closely monitored in all patients. ⁸⁰
Cytotoxic chemotherapy	Specialist/some GPs	Bone marrow suppression, GI upset, other	Indicated in lymphoma, not strongly recommended for TCC
Urethral stent ⁹²	Specialist	Infections and incontinence— $\leq 50\%$ ⁹²	Consider if obstructed—palliative only

Abbreviations: GI, gastrointestinal; GP, general practitioner; NSAID, nonsteroidal anti-inflammatory drug; PO, orally; SID, once a day; TCC, transitional cell carcinoma.

prognosticate for other bladder tumor types or for primary prostate tumors, but surgery should be considered where appropriate because prolonged, good outcomes have been reported in some cases.

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REFERENCES

1. Knapp DW, Ramos-Vara JA, Moore GE, et al. Urinary bladder cancer in dogs, a naturally occurring model for cancer biology and drug development. *ILAR J* 2014;55:100–18.
2. Knapp DW, Glickman NW, Denicola DB, et al. Naturally-occurring canine transitional cell carcinoma of the urinary bladder: a relevant model of human invasive bladder cancer. *Urol Oncol* 2000;5:47–59.
3. Knapp DW. Animal models of naturally occurring canine urinary bladder cancer. In: Lerner SP, Schoenberg MP, Sternberg C, editors. *Textbook of bladder cancer*. (United Kingdom): Taylor and Francis; 2006. p. 171–5.

4. Mutsaers AJ, Widmer WR, Knapp DW. Canine transitional cell carcinoma. *J Vet Intern Med* 2003;17(2):136–44.
5. Norris AM, Laing EJ, Valli VE, et al. Canine bladder and urethral tumors: a retrospective study of 115 cases (1980–1985). *J Vet Intern Med* 1992;6:149–53.
6. Hayes HM Jr. Canine bladder cancer: epidemiologic features. *Am J Epidemiol* 1976;104:673–7.
7. Knapp DW, McMillan SK. Tumors of the urinary system. In: Withrow SJ, Vail DM, Page RL, editors. *Withrow and MacEwen's small animal clinical oncology*. 5th edition. St Louis (MO): Saunders Elsevier; 2013. p. 572–82.
8. Glickman LT, Raghavan M, Knapp DW, et al. Herbicide exposure and the risk of transitional cell carcinoma of the urinary bladder in Scottish Terriers. *J Am Vet Med Assoc* 2004;224:1290–7.
9. Raghavan M, Knapp DW, Bonney PL, et al. Evaluation of the effect of dietary vegetable consumption on reducing risk of transitional cell carcinoma of the urinary bladder in Scottish Terriers. *J Am Vet Med Assoc* 2005;227:94–100.
10. Tarvin G, Patnaik A, Greene R. Primary urethral tumors in dogs. *J Am Vet Med Assoc* 1978;172:931–3.
11. Santos M, Dias Pereira P, Montenegro L, et al. Recurrent and metastatic canine urethral transitional cell carcinoma without bladder involvement. *Vet Rec* 2007;160:557–8.
12. Nieset JR, Harmon JF, Larue SM. Use of cone-beam computed tomography to characterize daily urinary bladder variations during fractionated radiotherapy for canine bladder cancer. *Vet Radiol Ultrasound* 2011;52:580–8.
13. Borjesson DL, Christopher MM, Ling GV. Detection of canine transitional cell carcinoma using a bladder tumor antigen urine dipstick test. *Vet Clin Pathol* 1999;28:33–8.
14. Henry CJ, Tyler JW, McEntee MC, et al. Evaluation of a bladder tumor antigen test as a screening test for transitional cell carcinoma of the lower urinary tract in dogs. *Am J Vet Res* 2003;64:1017–20.
15. Nyland TG, Wallack ST, Wisner ER. Needle-tract implantation following us-guided fine-needle aspiration biopsy of transitional cell carcinoma of the bladder, urethra, and prostate. *Vet Radiol Ultrasound* 2002;43:50–3.
16. Vignoli M, Rossi F, Chierici C, et al. Needle tract implantation after fine needle aspiration biopsy (FNAB) of transitional cell carcinoma of the urinary bladder and adenocarcinoma of the lung. *Schweiz Arch Tierheilkd* 2007;149:314–8.
17. Molnar T, Vajdovich P. Clinical factors determining the efficacy of urinary bladder tumour treatments in dogs: surgery, chemotherapy or both? *Acta Vet Hung* 2012;60:55–68.
18. Robat C, Burton J, Thamm D, et al. Retrospective evaluation of doxorubicin-piroxicam combination for the treatment of transitional cell carcinoma in dogs. *J Small Anim Pract* 2013;54:67–74.
19. Blackburn AL, Berent AC, Weisse CW, et al. Evaluation of outcome following urethral stent placement for the treatment of obstructive carcinoma of the urethra in dogs: 42 cases (2004–2008). *J Am Vet Med Assoc* 2013;242:59–68.
20. Weisse C, Berent A, Todd K, et al. Evaluation of palliative stenting for management of malignant urethral obstructions in dogs. *J Am Vet Med Assoc* 2006;229:226–34.
21. Cerf DJ, Lindquist EC. Palliative ultrasound-guided endoscopic diode laser ablation of transitional cell carcinomas of the lower urinary tract in dogs. *J Am Vet Med Assoc* 2012;240:51–60.

22. Upton ML, Tangner CH, Payton ME. Evaluation of carbon dioxide laser ablation combined with mitoxantrone and piroxicam treatment in dogs with transitional cell carcinoma. *J Am Vet Med Assoc* 2006;228:549–52.
23. Knapp DW, Henry CJ, Widmer WR, et al. Randomized trial of cisplatin versus firocoxib versus cisplatin/firocoxib in dogs with transitional cell carcinoma of the urinary bladder. *J Vet Intern Med* 2013;27:126–33.
24. Knapp DW, Richardson RC, Chan TC, et al. Piroxicam therapy in 34 dogs with transitional cell carcinoma of the urinary bladder. *J Vet Intern Med* 1994; 8:273–8.
25. McMillan SK, Boria P, Moore GE, et al. Antitumor effects of deracoxib treatment in 26 dogs with transitional cell carcinoma of the urinary bladder. *J Am Vet Med Assoc* 2011;239:1084–9.
26. Desai JC, Sanyai SM, Goo T, et al. Primary prevention of adverse gastroduodenal effects from short-term use of non-steroidal anti-inflammatory drugs by omeprazole 20mg in healthy subjects: a randomized, double-blind, placebo controlled study. *Dig Dis Sci* 2008;53:2059–65.
27. Graham DY, Agrawal NM, Campbell DR, et al. Ulcer prevention in long-term users of nonsteroidal anti-inflammatory drugs: results of a double-blind, randomized, multicenter active- and placebo-controlled study of misoprostol vs lansoprazole. *Arch Intern Med* 2002;162:169–75.
28. Cullen D, Bardhan KD, Eisner M, et al. Primary gastroduodenal prophylaxis with omeprazole for non-steroidal anti-inflammatory drug users. *Aliment Pharmacol Ther* 1998;12:135–40.
29. Henry CJ, McCaw DL, Turnquist SE, et al. Clinical evaluation of mitoxantrone and piroxicam in a canine model of human invasive urinary bladder carcinoma. *Clin Cancer Res* 2003;9:906–11.
30. Chun R, Knapp DW, Widmer WR, et al. Phase II clinical trial of carboplatin in canine transitional cell carcinoma of the urinary bladder. *J Vet Intern Med* 1997;11:279–83.
31. Boria PA, Glickman NW, Schmidt BR, et al. Carboplatin and piroxicam therapy in 31 dogs with transitional cell carcinoma of the urinary bladder. *Vet Comp Oncol* 2005;3:73–80.
32. Chun R, Knapp DW, Widmer WR, et al. Cisplatin treatment of transitional cell carcinoma of the urinary bladder in dogs: 18 cases (1983-1993). *J Am Vet Med Assoc* 1996;209:1588–91.
33. Knapp DW, Glickman NW, Widmer WR, et al. Cisplatin versus cisplatin combined with piroxicam in a canine model of human invasive urinary bladder cancer. *Cancer Chemother Pharmacol* 2000;46:221–6.
34. Greene SN, Lucroy MD, Greenberg CB, et al. Evaluation of cisplatin administered with piroxicam in dogs with transitional cell carcinoma of the urinary bladder. *J Am Vet Med Assoc* 2007;231:1056–60.
35. Arnold EJ, Childress MO, Fourez LM, et al. Clinical trial of vinblastine in dogs with transitional cell carcinoma of the urinary bladder. *J Vet Intern Med* 2011; 25:1385–90.
36. Marconato L, Zini E, Lindner D, et al. Toxic effects and antitumor response of gemcitabine in combination with piroxicam treatment in dogs with transitional cell carcinoma of the urinary bladder. *J Am Vet Med Assoc* 2011;238: 1004–10.
37. Kaye ME, Thamm DH, Weishaar K, et al. Vinorelbine rescue therapy for dogs with primary urinary bladder carcinoma. *Vet Comp Oncol* 2013. <http://dx.doi.org/10.1111/vco.12065>.

38. Schrempp DR, Childress MO, Stewart JC, et al. Metronomic administration of chlorambucil for treatment of dogs with urinary bladder transitional cell carcinoma. *J Am Vet Med Assoc* 2013;242:1534–8.
39. Abbo AH, Jones DR, Masters AR, et al. Phase I clinical trial and pharmacokinetics of intravesical mitomycin C in dogs with localized transitional cell carcinoma of the urinary bladder. *J Vet Intern Med* 2010;24:1124–30.
40. Foroudi F, Pham D, Bressel M, et al. Bladder cancer radiotherapy margins: a comparison of daily alignment using skin, bone or soft tissue. *Clin Oncol* 2012;24:673–81.
41. Ichinohe K, Ijima M, Usami T, et al. Complete remission of primary retroperitoneal transitional cell carcinoma after radiotherapy and oral chemotherapy: a case report. *Ann R Coll Surg Engl* 2013;95:e52–4.
42. Jang NY, Kim IA, Byun SS, et al. Patterns of failure and prognostic factors for locoregional recurrence after radical surgery in upper urinary tract transitional cell carcinoma: implications for adjuvant radiotherapy. *Urol Int* 2013;90:202–6.
43. Mitin T, Hunt D, Shipley WU, et al. Transurethral surgery and twice-daily radiation plus paclitaxel-cisplatin or fluorouracil-cisplatin with selective bladder preservation and adjuvant chemotherapy for patients with muscle invasive bladder cancer (RTOG 0233): a randomised multicentre phase 2 trial. *Lancet Oncol* 2013;14:863–72.
44. Zapatero A, Martin De Vidales C, Arellano R, et al. Long-term results of two prospective bladder-sparing trimodality approaches for invasive bladder cancer: neoadjuvant chemotherapy and concurrent radio-chemotherapy. *Urology* 2012;80:1056–62.
45. Walker MB, Breider M. Intraoperative radiotherapy of canine bladder cancer. *Vet Radiol* 1987;28:200–4.
46. Poirier VJ, Forrest LJ, Adams WM, et al. Piroxicam, mitoxantrone, and coarse fraction radiotherapy for the treatment of transitional cell carcinoma of the bladder in 10 dogs: a pilot study. *J Am Anim Hosp Assoc* 2004;40:131–6.
47. Nolan MW, Kogan L, Griffin LR, et al. Intensity-modulated and image-guided radiation therapy for treatment of genitourinary carcinomas in dogs. *J Vet Intern Med* 2012;26:987–95.
48. Bryan JN, Keeler MR, Henry CJ, et al. A population study of neutering status as a risk factor for canine prostate cancer. *Prostate* 2007;67:1174–81.
49. Assin R, Baldi A, Citro G, et al. Prostate as sole unusual recurrence site of lymphoma in a dog. *In Vivo* 2008;22:755–7.
50. Winter MD, Locke JE, Penninck DG. Imaging diagnosis—urinary obstruction secondary to prostatic lymphoma in a young dog. *Vet Radiol Ultrasound* 2006;47:597–601.
51. Della Santa D, Dandrieux J, Psalla D, et al. Primary prostatic haemangiosarcoma causing severe haematuria in a dog. *J Small Anim Pract* 2008;49:249–51.
52. Hayden DW, Bartges JW, Bell FW, et al. Prostatic hemangiosarcoma in a dog: clinical and pathologic findings. *J Vet Diagn Invest* 1992;4:209–11.
53. Hayden DW, Klausner JS, Waters DJ. Prostatic leiomyosarcoma in a dog. *J Vet Diag Invest* 1999;11:283–6.
54. Krawiec DR, Heflin D. Study of prostatic disease in dogs: 177 cases (1981–1986). *J Am Vet Med Assoc* 1992;200:1119–22.
55. Cornell KK, Bostwick DG, Cooley DM, et al. Clinical and pathologic aspects of spontaneous canine prostate carcinoma: a retrospective analysis of 76 cases. *Prostate* 2000;45:173–83.

56. Bell FW, Klausner JS, Hayden DW, et al. Clinical and pathologic features of prostatic adenocarcinoma in sexually intact and castrated dogs: 31 cases (1970-1987). *J Am Vet Med Assoc* 1991;199:1623-30.
57. Malek S, Murphy KA, Nykamp SG, et al. Metastatic transitional cell carcinoma in proximal humerus of a dog. *Can Vet J* 2011;52:1013-7.
58. Cooley DM, Waters DJ. Skeletal metastasis as the initial clinical manifestation of metastatic carcinoma in 19 dogs. *J Vet Intern Med* 1998;12:288-93.
59. Bradbury CA, Westropp JL, Pollard RE. Relationship between prostatomegaly, prostatic mineralization, and cytologic diagnosis. *Vet Radiol Ultrasound* 2009;50:167-71.
60. Feeney DA, Johnston GR, Klausner JS, et al. Canine prostatic disease—comparison of radiographic appearance with morphologic and microbiologic findings: 30 cases (1981-1985). *J Am Vet Med Assoc* 1987;190:1018-26.
61. Feeney DA, Johnston GR, Klausner JS, et al. Canine prostatic disease—comparison of ultrasonographic appearance with morphologic and microbiologic findings: 30 cases (1981-1985). *J Am Vet Med Assoc* 1987;190:1027-34.
62. Swinney GR. Prostatic neoplasia in five dogs. *Aust Vet J* 1998;76:669-74.
63. Higuchi T, Burcham GN, Childress MO, et al. Characterization and treatment of transitional cell carcinoma of the abdominal wall in dogs: 24 cases (1985-2010). *J Am Vet Med Assoc* 2013;242:499-506.
64. Mohammed SI, Khan KN, Sellers RS, et al. Expression of cyclooxygenase-1 and 2 in naturally-occurring canine cancer. *Prostaglandins Leukot Essent Fatty Acids* 2004;70:479-83.
65. L'Eplattenier HF, Lai CL, van den Ham R, et al. Regulation of COX-2 expression in canine prostate carcinoma: increased COX-2 expression is not related to inflammation. *J Vet Intern Med* 2007;21:776-82.
66. Sorenmo KU, Goldschmidt MH, Shofer FS, et al. Evaluation of cyclooxygenase-1 and cyclooxygenase-2 expression and the effect of cyclooxygenase inhibitors in canine prostatic carcinoma. *Vet Comp Oncol* 2004;2:13-23.
67. Axiak-Bechtel SM, Kumar SR, Dank KK, et al. Nanoparticulate paclitaxel demonstrates antitumor activity in PC3 and Ace-1 aggressive prostate cancer cell lines. *Invest New Drugs* 2013;31:1609-15.
68. Osborne CA, Low DG, Perman V, et al. Neoplasms of the canine and feline urinary bladder: incidence, etiologic factors, occurrence and pathologic features. *Am J Vet Res* 1968;29:2041-55.
69. Barrett RE, Nobel TA. Transitional cell carcinoma of the urethra in a cat. *Cornell Vet* 1976;66:14-26.
70. Caney SM, Holt PE, Day MJ, et al. Prostatic carcinoma in two cats. *J Small Anim Pract* 1998;39:140-3.
71. Tursi M, Costa T, Valenza F, et al. Adenocarcinoma of the disseminated prostate in a cat. *J Feline Med Surg* 2008;10:600-2.
72. Zambelli D, Cunto M, Raccagni R, et al. Successful surgical treatment of a prostatic biphasic tumour (sarcomatoid carcinoma) in a cat. *J Feline Med Surg* 2010;12:161-5.
73. Hubbard BS, Vulgamott JC, Liska WD. Prostatic adenocarcinoma in a cat. *J Am Vet Med Assoc* 1990;197:1493-4.
74. LeRoy BE, Lech ME. Prostatic carcinoma causing urethral obstruction and obstipation in a cat. *J Feline Med Surg* 2004;6:397-400.
75. Patnaik AK, Schwarz PD, Greene RW. A histopathologic study of twenty urinary bladder neoplasms in the cat. *J Small Anim Pract* 1986;27:433-45.

76. Wilson HM, Chun R, Larson VS, et al. Clinical signs, treatments, and outcome in cats with transitional cell carcinoma of the urinary bladder: 20 cases (1990-2004). *J Am Vet Med Assoc* 2007;231:101-6.
77. Simpson CJ, Mansfield CS, Milne ME, et al. Central diabetes insipidus in a cat with central nervous system B cell lymphoma. *J Feline Med Surg* 2011;13:787-92.
78. Schwarz PD, Greene RW, Patnaik AK. Urinary-bladder tumors in the cat - a review of 27 cases. *J Am Anim Hosp Assoc* 1985;21:237-45.
79. Brearley MJ, Thatcher C, Cooper JE. Three cases of transitional cell carcinoma in the cat and a review of the literature. *Vet Rec* 1986;118:91-4.
80. Bommer NX, Hayes AM, Scase TJ, et al. Clinical features, survival times and COX-1 and COX-2 expression in cats with transitional cell carcinoma of the urinary bladder treated with meloxicam. *J Feline Med Surg* 2012;14:527-33.
81. Burk RL, Meierhenry EF, Schaubhut CW Jr. Leiomyosarcoma of the urinary bladder in a cat. *J Am Vet Med Assoc* 1975;167:749-51.
82. Pavia PR, Havig ME, Donovan TA, et al. Malignant peripheral nerve sheath tumour of the urinary bladder in a cat. *J Small Anim Pract* 2012;53:245-8.
83. Patnaik AK, Greene RW. Intravenous leiomyoma of the bladder in a cat. *J Am Vet Med Assoc* 1979;175:381-3.
84. Khodakaram-Tafti A, Shirian S, Vesal N, et al. Lipoma of the urinary bladder in a cat. *J Comp Pathol* 2011;144:212-3.
85. Benigni L, Lamb CR, Corzo-Menendez N, et al. Lymphoma affecting the urinary bladder in three dogs and a cat. *Vet Radiol Ultrasound* 2006;47:592-6.
86. Geigy CA, Dandrieux J, Miclard J, et al. Extranodal B-cell lymphoma in the urinary bladder with cytological evidence of concurrent involvement of the gall bladder in a cat. *J Small Anim Pract* 2010;51:280-7.
87. Sutherland-Smith M, Harvey C, Campbell M, et al. Transitional cell carcinomas in four fishing cats (*Prionailurus viverrinus*). *J Zoo Wildl Med* 2004;35:370-80.
88. Landolfi JA, Terio KA. Transitional cell carcinoma in fishing cats (*Prionailurus viverrinus*): pathology and expression of cyclooxygenase-1, -2, and p53. *Vet Pathol* 2006;43:674-81.
89. Buffington CA, Chew DJ, Kendall MS, et al. Clinical evaluation of cats with non-obstructive urinary tract diseases. *J Am Vet Med Assoc* 1997;210:46-50.
90. Sellon RK, Rottman JB, Jordan HL, et al. Hypereosinophilia associated with transitional cell carcinoma in a cat. *J Am Vet Med Assoc* 1992;201:591-3.
91. Walker DB, Cowell RL, Clinkenbeard KD, et al. Carcinoma in the urinary bladder of a cat: cytologic findings and a review of the literature. *Vet Clin Pathol* 1993;22:103-8.
92. Brace MA, Weisse C, Berent A. Preliminary experience with stenting for management of non-urolith urethral obstruction in eight cats. *Vet Surg* 2014;43:199-208.
93. Newman RG, Mehler SJ, Kitchell BE, et al. Use of a balloon-expandable metallic stent to relieve malignant urethral obstruction in a cat. *J Am Vet Med Assoc* 2009;234:236-9.
94. Lucroy MD, Bowles MH, Higbee RG, et al. Photodynamic therapy for prostatic carcinoma in a dog. *J Vet Intern Med* 2003;17:235-7.
95. L'Epattenier HF, Klem B, Teske E, et al. Preliminary results of intraoperative photodynamic therapy with 5-aminolevulinic acid in dogs with prostate carcinoma. *Vet J* 2008;178:202-7.
96. Goldsmid SE, Bellenger CR. Urinary incontinence after prostatectomy in dogs. *Vet Surg* 1991;20:253-6.

97. L'Eplattenier HF, van Nimwegen SA, van Sluijs FJ, et al. Partial prostatectomy using Nd:YAG laser for management of canine prostate carcinoma. *Vet Surg* 2006;35:406–11.
98. Vlasin M, Rauser P, Fichtel T, et al. Subtotal intracapsular prostatectomy as a useful treatment for advanced-stage prostatic malignancies. *J Small Anim Pract* 2006;47:512–6.
99. Sorenmo KU, Goldschmidt M, Shofer F, et al. Immunohistochemical characterization of canine prostatic carcinoma and correlation with castration status and castration time. *Vet Comp Oncol* 2003;1:48–56.
100. Teske E, Naan EC, van Dijk EM, et al. Canine prostate carcinoma: epidemiological evidence of an increased risk in castrated dogs. *Mol Cell Endocrinol* 2002;197:251–5.
101. Obradovich J, Walshaw R, Goullaud E. The influence of castration on the development of prostatic carcinoma in the dog 43 cases (1978-1985). *J Vet Intern Med* 1987;1:183–7.
102. Lai CL, L'Eplattenier H, van den Ham R, et al. Androgen receptor CAG repeat polymorphisms in canine prostate cancer. *J Vet Intern Med* 2008;22:1380–4.